Michael Mensah

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EDUCATION

The Pennsylvania State University | ***College of Engineering*** **GPA: 3.66**
B.S. in Computer Science, Minor in Entrepreneurship and Innovation Expected: May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Design & Organization, Communication Networks

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL
Frameworks: React, Next.js, Node.js, Express, FastAPI
Data & Software Engineering: PostgreSQL, REST APIs, Git, GitHub Actions, Automated Testing, Debugging, CI/CD
Cloud & Infrastructure: IBM Cloud, Docker, Kubernetes, OpenShift, Helm, Linux
AI/ML: RAG, Vector Search, LLM Tool Calling, MCP, Evaluation & Benchmarking

EXPERIENCE

IBM | Customer Success Engineer Co-op | AI & Automation **May 2025 – Present**

- Built and shipped a production Python retrieval service across 2,300+ governance documents, reducing customer-proposal first-draft time 86% from 3 hours to 25 minutes for 12 pilot users.
- Validated retrieval against a 1,400-question regression benchmark, reaching 93% top-3 accuracy at 120ms latency; added PII redaction and domain routing before production rollout.
- Built API and token endpoints for an enterprise catalog of 1,700+ demos, enabling programmatic discovery and AI application integration via MCP.
- Automated a 6-step weekly manual workflow into a Helm-managed Kubernetes CronJob syncing analytics across the full demo catalog, with retry and concurrency controls.
- Deployed a containerized GenAI microservice on IBM Cloud using Docker, Kubernetes, and OpenShift, reducing artifact-triage time 55%.

PROJECTS

AI Movie Concierge | TypeScript, React, Node.js, Express, MCP | Live | Code **Oct 2026**

- Built and deployed a TypeScript application integrating TMDB and Letterboxd through a custom 8-tool MCP server, delivering personalized movie recommendations with live agent streaming.
- Designed "Blend," using cosine similarity over rating-weighted genre vectors to compare two users' tastes and recommend unseen films tailored to both.
- Achieved a 100% pass rate across 21 TMDB ground-truth evaluation cases with code-enforced grounding, backed by 55 automated tests in GitHub Actions CI.

Full-Stack Onboarding Platform | TypeScript, Next.js, FastAPI, PostgreSQL **May 2026 – Present**

- Built and shipped a full-stack onboarding platform adopted by 30+ IBM interns and managers, combining daily streaks, XP, badges, team management, and manager grading in a single application.
- Owned production debugging and deployment, shipping same-day fixes for a PostgREST data-loss bug and Next.js asset-serving failure that restored streak tracking and core application functionality.

AI Developer Tool | TypeScript, Node.js, MCP **Jun 2026**

- Led a 6-person team to 1st place out of 8 teams, building a TypeScript and Node.js developer tool that gives AI coding assistants file-level design context from a single click.
- Built an MCP server with allowlisted file-access guardrails, restricting AI-assisted edits to approved files and keeping generated changes scoped and reviewable.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow (MLT) | *Career Prep Fellow* **Feb 2024 – Present**

- Selected for a highly competitive 20-month professional development fellowship with technical, business, and leadership training from industry mentors.

A. James Clark Scholars Program | *Scholar & Mentor* **Apr 2023 – Present**

- Awarded a full-ride scholarship given to 10 College of Engineering students each year; mentor a cohort of 40 engineering scholars.

Penn State Engineering Ambassadors | *Engineering Ambassador* **May 2025 – Present**

- Coordinated 5 campus STEM days; managed 14 volunteers and logistics end-to-end, earning 95% teacher satisfaction.